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Corallocins A−C, Nerve Growth and Brain-Derived Neurotrophic Factor Inducing Metabolites from the Mushroom *Hericium coralloides*

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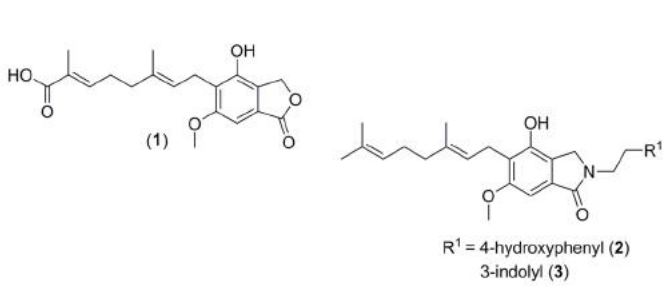
Supporting Information



ABSTRACT: Three new natural products, corallocins A−C (**1**−**3**), along with two known compounds were isolated from the mushroom *Hericium coralloides*. Their benzofuranone and isoindolinone structures were elucidated by spectral methods. All corallocins induced nerve growth factor and/or brain-derived neurotrophic factor expression in human 1321N1 astrocytes. Furthermore, corallocin B showed antiproliferative activity against HUVEC and human cancer cell lines MCF-7 and KB-3-1.

Hericium coralloides, a basidiomycete with peculiar fruiting bodies resembling white coral, belongs to the family Hericiaceae. The most famous representative of this family, *H. erinaceus*, has been used in traditional Chinese medicine for a long time and is processed into food supplements and alternative medicines.[1−3] Numerous publications report various biological activities of its secondary metabolites such as antibacterial, cytotoxic, and neuritogenic effects.[4−9] In contrast, there are only a few reports of the metabolite production of *H. coralloides* (also referred to under its synonyms *H. ramosum* and *H. abietinum*).

In the course of a study on natural products with neurotrophic effects, three new metabolites (**1**−**3**), including an unprecedented indole isoindolinone derivative (**3**), with nerve growth factor (NGF)- and brain-derived neurotrophic factor (BDNF)-inducing activities, were obtained from fruiting bodies of *H. coralloides* together with hericerin (**4**)[10] and an isoindolinone derivative (**5**),[11] known from *H. erinaceus* (Figure 1). NGF and BDNF are two important factors of the nervous system, regulating growth, development, and survival of neurons.[12−14] Since deficiency of neurotrophic factors is related to different neurodegenerative diseases including Alzheimer's disease,[15,16] research has focused on the therapeutic potential of small molecules with neurotrophin-enhancing properties. Several publications report NGF-inducing natural products isolated from different sources[17−20] as well as synthetic approaches.[21,22] Recently, it has been shown that hericenones and erinacines, isolated from fruiting bodies and mycelium of *H. erinaceus*, also promote NGF biosynthesis in mouse astroglial cells in vitro[5,23,24] and in vivo.[25] Corallocins A−C (**1**−**3**) were found to induce different patterns of neurotrophin expression in 1321N1 astrocytes, indicating different molecular targets. Consequently, treatment of neural PC12 cells with conditioned media produced by these astrocytoma cells resulted in different degrees of differentiation. In addition, corallocin B (**2**) showed an antiproliferative effect on human endothelial cells HUVEC (IC_{50} 2.1 μM) and human cancer cells MCF-7 (IC_{50} 9.2 μM) and KB-3-1 (IC_{50} 11.5 μM).

RESULTS AND DISCUSSION

Basidiomes (fruiting bodies; 2.8 kg) of *H. coralloides* were homogenized and extracted with acetone (4 L) for 48 h. After evaporation of the solvent the remaining aqueous residue was extracted with ethyl acetate (1 L) and the crude extract (7.88 g) was subjected to silica flash chromatography (petroleum ether−ethyl acetate, 1:0 to 0:1). The resulting fractions (I−IV) were purified by preparative HPLC [RP-C_{18}, water−acetonitrile + 0.05% trifluoroacetic acid (TFA)] to afford compounds **1**−**5**. Their structures were elucidated by ESIMS and interpretation of extensive NMR data.

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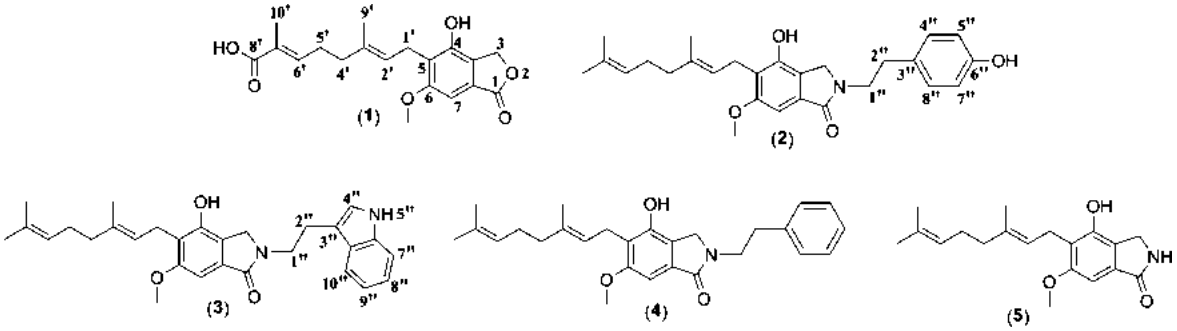

Figure 1. Compounds (**1**−**5**) isolated from basidiomes of *H. coralloides*.

Table 1. ^{1}H and ^{13}C NMR Data (500 MHz, 175 MHz) of Corallocins A−C (**1**−**3**) in Acetone-d_6

	1		2		3	
position	δ_C, type	δ_H, m (J in Hz)	δ_C, type	δ_H, m (J in Hz)	δ_C, type	δ_H, m (J in Hz)
1	171.7, C		168.8, C		168.7, C	
2					121.8, N	
3	68.7, CH_2	5.25, s	48.9, CH_2	4.24, s	48.6, CH_2	4.29, s
3a	125.6, C		121.9, C		121.9, C	
4	150.5, C		150.8, C		150.8, C	
5	124.1, C		121.2, C		121.0, C	
6	160.7, C		159.9, C		159.9, C	
7	98.7, CH	6.89, s	97.8, CH	6.83, s	97.8, CH	6.85, s
7a	127.7, C		133.0, C		133.3, C	
OMe	56.5, CH_3	3.91, s	56.4, CH_3	3.86, s	56.4, CH_3	3.88, s
1′	23.5, CH_2	3.47, d (7.2)	23.3, CH_2	3.43, d (7.2)	23.3, CH_2	3.42, d (7.2)
2′	123.5, CH	5.26, m	123.4, CH	5.21, tq (7.2, 1.3)	123.4, CH	5.20, tq (7.2, 1.3)
3′	135.2, C		135.5, C		135.5, C	
4′	39.0, CH_2	2.1, t (7.4)	40.6, CH_2	1.94, m	40.6, CH_2	1.94, m
5′	27.8, CH_2	2.28, m	27.5, CH_2	2.02, m	27.5, CH_2	2.03, m
6′	142.6, CH	6.71, tq (7.4, 1.4)	125.2, CH	5.05, m	125.2, CH	5.06, m
7′	128.4, C		131.7, C		131.7, C	
8′	169.1, C		25.9, CH_3	1.60, d (1.1)	25.9, CH_3	1.60, d (1.2)
9′	16.2, CH_3	1.81, d (1.2)	16.3, CH_3	1.77, d (1.3)	16.3, CH_3	1.77, d (1.3)
10′	12.5, CH_3	1.75, m	17.8, CH_3	1.54, s	17.8, CH_3	1.54, s
1″			45.0, CH_2	3.76, t (7.3)	43.6, CH_2	3.90, t (7.3)
2″			34.7, CH_2	2.87, t (7.3)	25.3, CH_2	3.12, t (7.3)
3″			130.9, C		113.1, C	
4″			130.6, CH	7.07, d (8.5)	123.3, CH	7.18, s
5″			116.2, CH	6.75, d (8.5)	125.5, NH	10.0, s br
6″			156.8, C		137.7, C	
7″			116.2, CH	6.75, d (8.5)	112.2, CH	7.38, ddd (8.1, 0.9, 0.9)
8″			130.6, CH	7.07, d (8.5)	122.3, CH	7.10, ddd (8.1, 7.0, 1.1)
9″					119.6, CH	7.02, ddd (7.9, 7.0, 1.0)
10″					119.3, CH	7.65, ddd (7.8, 1.1, 0.9)
11″					128.6, C	

Corallocin A (**1**) was obtained as a yellowish oil and displayed two major peaks in the HR-ESIMS spectrum at m/z = 347.1490 ([M + H]$^+$, calcd for $C_{19}H_{23}O_6$ 347.1489) and at m/z = 693.2912 ([2M + H]$^+$, calcd for $C_{38}H_{45}O_{12}$ 693.2905), consistent with the molecular formula $C_{19}H_{22}O_6$ and indicating 9 degrees of unsaturation. The IR spectrum showed absorption bands at 3410, 2960, and 1685 cm^{-1}, revealing the presence of hydroxy and carbonyl groups. The ^{1}H NMR spectrum of **1** (Table 1) exhibited two methyl signals at δ_H 1.75 (3H, m) and 1.81 (3*H*, d, J = 1.2 Hz), a singlet at δ_H 3.91 (3H, s) characteristic of an aromatic methoxy group, three methylene groups at δ_H 2.1 (2H, t, J = 7.4 Hz), 2.28 (2H, m) and 3.47 (2H, d, J = 7.2 Hz), overlapping signals around δ_H 5.26 of three protons, and two methines at δ_H 6.71 (1H, tq, J = 7.4, 1.4 Hz) and 6.89 (1H, s). The ^{13}C NMR and HSQC-DEPT spectra implied the presence of nine non-proton-bearing carbons with shifts in the range δ_C 172−124 ppm, including two carbonyl carbons (δ_C 171.7, 169.1). In addition, the analysis uncovered a fourth methylene group at δ_H 5.25 (s, overlapping signals) with the corresponding carbon at δ_C 68.7. This downfield shift suggested a position adjacent to an oxygen. On the basis of COSY and HMBC correlations an unsaturated alkyl chain was established (Figure 2). The chemical shift of carbonyl C-8′ (δ_C 169.1) and the HMBC correlations of H-6′ (δ_H 6.71) and CH_3-10′ (δ_H 1.75) to this carbon indicated a carboxylic acid moiety at the end of the alkyl chain. The remaining part of **1** was

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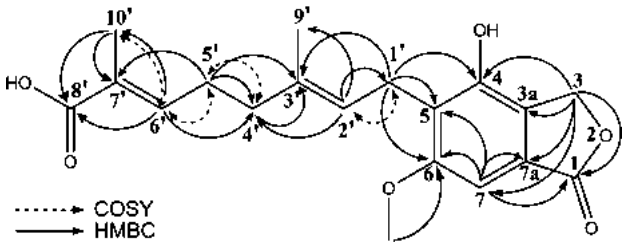

Figure 2. Key ^{1}H−^{1}H COSY and ^{1}H−^{13}C HMBC correlations of corallocin A (**1**).

identified as 2-benzofuranone by detailed analysis of all further correlations. For methylene $\delta_{C\text{-}3}$ 68.7 ($\delta_{H\text{-}3}$ 5.25) HMBC correlations to the second carbonyl carbon $\delta_{C\text{-}1}$ 171.7 were detected as well as to the aromatic non-proton-bearing carbons $\delta_{C\text{-}3a}$ 125.6, $\delta_{C\text{-}7a}$ 127.7, and $\delta_{C\text{-}4}$ 150.5 and the aromatic proton $\delta_{C\text{-}7}$ 98.7 ($\delta_{H\text{-}7}$ 6.89). Finally, HMBC correlations of the methoxy group δ_H 3.91 and the alkyl chain methylene $\delta_{H\text{-}1'}$ 3.47 confirmed the position of the substituents and unraveled the structure of corallocin A (**1**). It shares the same 4-hydroxy-6-methoxy-1(3*H*)-isobenzofuranone fragment with erinacerin B,[26] isolated from *H. erinaceus*, but shows differences in the alkyl chain.

HR-ESIMS of corallocin B (**2**) revealed the presence of nitrogen. The spectrum exhibited a $[M + H]^+$ peak at m/z = 436.2484 (calcd for $C_{27}H_{34}NO_4$ m/z = 436.2482), consistent with the molecular formula $C_{27}H_{33}NO_4$. The IR spectrum displayed an absorption at 1660 cm^{-1}, indicating a carbonyl group (or amide moiety), and prominent absorption bands at 3280 (broad) and 2935 cm^{-1}, which are characteristic of hydroxy groups. The ^{1}H NMR data of **2** indicated three methyl groups at δ_H 1.54 (3H, s), 1.60 (3H, d, J = 1.1 Hz), and 1.77 (3H, d, J = 1.3 Hz) and a singlet at δ_H 3.86 (3H, s), displaying an aromatic methoxy group. Moreover, the spectrum showed six methylene groups with chemical shifts varying from 1.94 to 4.24 ppm, two olefinic methines with complex spin systems at δ_H 5.05 (1H, m) and 5.21 (1H, tq, J = 7.2, 1.3 Hz), and three resonances of aromatic protons. In addition to one singlet resonance at δ_H 6.83 (1H, s), two doublets of triplets were detected in the aromatic region, representing two protons each. The ^{1}H−^{1}H COSY experiment confirmed the correlation between these four protons, and HSQC-DEPT revealed two symmetric protons at $\delta_{H\text{-}5''/7''}$ 6.75 (2H, d, J = 8.5 Hz) and $\delta_{H\text{-}4''/8''}$ 7.07 (2H, d, J = 8.5 Hz) with corresponding carbons at $\delta_{C\text{-}5''/7''}$ 116.2 and $\delta_{C\text{-}4''/8''}$ 130.6, typical of a 1,4-substituted benzene. The rather low chemical shift of the carbons $C_{5''/7''}$ with δ_C 116.2 ppm pointed to a hydroxy or amino substitution. HMBC correlations of these aromatic protons to the highly shifted carbons at $\delta_{C\text{-}6''}$ 156.8 and $\delta_{C\text{-}3''}$ 130.9 completed the identification of a phenylethyl substituent. Similar to **1**, the analysis of COSY and HMBC correlations resulted in an alkyl chain with two unsaturated bonds at C-2′−C-3′ and C-6′−C-7′, but with two tertiary methyl groups, $\delta_{H\text{-}8'}$ 1.60 and $\delta_{H\text{-}10'}$ 1.54, at the end of the chain and one tertiary methyl $\delta_{H\text{-}9'}$ 1.77 at position C-3′, leading to the 3,7-dimethylocta-2,6-dien-1-yl fragment. The downfield shift of the methylene singlet $\delta_{H\text{-}3}$ 4.24 (2H, s, δ_C 48.9) was not as strong as in corallocin A with $\delta_{H\text{-}3}$ 5.25 ($\delta_{C\text{-}3}$ 68.7), suggesting the presence of an adjacent nitrogen. Otherwise the chemical shifts and correlation pattern were comparable to **1**, indicating a 4-hydroxy-5-(3′,7′-dimethylocta-2′,6′-dien-1′-yl)-6-methoxyisoindolinone structure. Long-range correlations from methylene at $\delta_{H\text{-}1''}$ 3.76 ($\delta_{C\text{-}1''}$ 45.0) to the carbon signals $\delta_{C\text{-}3}$ 48.9, $\delta_{C\text{-}1}$ 168.8, $\delta_{C\text{-}2''}$ 34.7, and $\delta_{C\text{-}3''}$ 130.9 as well as a weak proton−proton correlation of $\delta_{H\text{-}1''}$ 3.76 to $\delta_{H\text{-}3}$ 4.24 linked phenylethyl and isoindolinone fragments, completing the structure of **2** (Figure 1).

Corallocin C (**3**) was isolated as a colorless oil. The $[M + H]^+$ peak at m/z = 459.2640 (calcd for $C_{29}H_{35}N_2O_3$: m/z = 459.2642) in the HR-ESIMS spectrum provided the molecular formula $C_{29}H_{34}N_2O_3$ with 14 degrees of unsaturation. The IR spectrum exhibited prominent, sharp absorptions at 1675, 2935, and 3410 cm^{-1}, implying carbonyl, hydroxy, and amino moieties. ^{1}H NMR and ^{13}C NMR spectra of compound **3** displayed many signals, which were identical or almost identical with the signals of **2** (Table 1). Differences were obvious only in the area of aromatic protons with six resonances in total and the methylene groups at positions C-1″ and C-2″, showing stronger downfield shifts than in **2**. Additionally, there was a broad signal of one proton at δ_H 10.0 (1H, s br). These findings indicated the same isoindolinone structure as corallocin B (**2**) with different substitution at N-2, including a second nitrogen atom.

Detailed analysis of the 2D NMR spectra (HSQC-DEPT, COSY, HMBC) of **3** revealed a proton−proton correlation from the methylene group at position C-2″ to an aromatic proton at δ_H 7.18 (1H, s) and long-range correlations to non-proton-bearing sp^2 carbons at δ_C 113.1 and 128.6 (Figure 3). For the aromatic proton at δ_H 7.18 only J-couplings to methylene C-2″ and to non-proton-bearing carbons at δ_C 113.1, 128.6, and 137.7 were observed, while COSY correlations indicated the connection of all remaining protons at δ_H 7.02 (1H, ddd, J = 7.9, 7.0, 1.0 Hz), 7.10 (1H, ddd, J = 8.1, 7.0, 1.1 Hz), 7.38 (1H, dd, J = 8.1, 0.9, 0.9 Hz), and 7.65 (1H, ddd, J = 7.8, 1.1, 0.9 Hz) within one aromatic ring. Finally, we established an unprecedented indole moiety as a substituent of N-2, which was confirmed by ^{1}H−^{15}N correlations (NH_3 was used for calibration). ^{1}H−^{15}N-HSQC allowed the assignment of the indole NH at $\delta_{H\text{-}5''}$ 10.0 with a corresponding signal for nitrogen at $\delta_{N\text{-}5''}$ 125.5 ppm. Long-range correlations from the

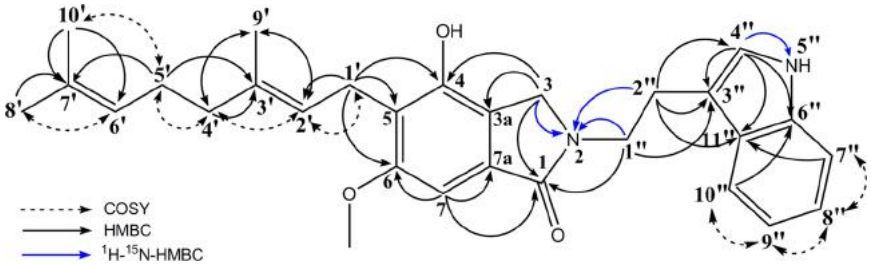

Figure 3. Key ^{1}H−^{1}H COSY and ^{1}H−^{13}C/^{15}N HMBC correlations of corallocin C (**1**).

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proton $\delta_{H\text{-}4''}$ 7.18 to the indole nitrogen $\delta_{N\text{-}5''}$ 125.5 and from the methylene groups at C-1″, C-2″, and C-3 to the second nitrogen at $\delta_{N\text{-}2}$ 121.8 further supported the structure of **3** as 5-[3,7-dimethylocta-2,6-dien-1-yl]-4-hydroxy-2-[2-(1*H*-indol-3-yl)ethyl]6-methoxy-2,3-dihydro-1*H*-isoindol-1-one. Compound **4** was identified as hericerin,[10] also referred to as isohericerin in several publication,[27,28] and **5** as [5-(2*E*)-3′,7′-dimethyl-2′,6′-octadienyl]-4-hydroxy-6-methoxy-1-isoindoline,[11] respectively, by comparing the NMR data with literature values (Figure 1).[11,29]

Compounds **1**−**5** were evaluated for cytotoxic and antimicrobial activities according to established procedures.[30] Only corallocin B (**2**) weakly inhibited growth of mouse fibroblast cells (L929), with an IC_{50} value of 14.7 μM, and therefore was tested against several human cell lines. It showed cytotoxic activity against HUVEC (IC_{50} 2.1 μM), MCF-7 (IC_{50} 9.2 μM), and KB-3-1 cells (IC_{50} 11.5 μM). Furthermore, **2** exhibited weak antifungal activity against *Mucor plumbeus* MUCL 49355 (MIC 57.4 μM).

After determining toxicity thresholds for compounds **1**−**5** also on 1321N1 astrocyte cells (Figure S1), the potential of corallocins A−C (**1**−**3**) to induce PC12 neuronal cell differentiation was analyzed. For PC12 and 1321N1 cell stimulation assays the highest nontoxic concentrations (no cell death after 24 h of incubation) were used [57.8 μM (**1**), 68.9 μM (**2**), and 19.6 μM (**3**)]. Upon addition of NGFβ-containing mouse salivary gland protein extracts (positive control), PC12 cells started to differentiate,[31] indicated by the outgrowth of neurites (Figure 4A, upper panel).

None of the tested compounds showed intrinsic neurotrophic activity, stimulating neurite outgrowth directly from cultured PC12 cells. However, two of the substances, **1** and **3**, significantly increased NGF secretion from 1321N1 astrocytes. Consequently, using conditioned 1321N1 culture medium, PC12 cells could be stimulated to differentiate (Figure 4B) compared to untreated controls. Corallocin C (**3**) displayed the highest activity (lower right panel) for stimulating neurite outgrowth from PC12 cells, inducing longer neurites compared to **1** (lower left panel), whereas compound **2** failed to stimulate differentiation of PC12 cells. A quantification of this assay is given in Figure 4C and Figure S2 (Supporting Information). These findings suggested that **3** induces higher levels of NGF expression compared to the other *Hericium* compounds, which was addressed by isolating mRNA from corallocin **1**−**3**-treated 1321N1 astrocytes for semiquantitative RT-PCR. Ubiquitously expressed glyceraldehyde 3-phosphate dehydrogenase (GAPDH) served as reference for calibration. Consistent with the neurite outgrowth analysis, NGF expression was upregulated in 1321N1 astrocytes upon stimulation by compounds **1** and **3** (Figure 5), with **3** being the strongest NGF inducer (about 5-fold; see Figure S3A) followed by **1** resulting in an approximately 3-fold increase and **2** in a 2-fold increase in NGF-mRNA compared to DMSO-treated controls. In parallel, we performed semiquantitative RT-PCR analysis for mRNA expression of BDNF. It does not induce neurite outgrowth from PC12 cells,[32] but stimulates neurogenesis and acts as a survival factor on neurons via the tropomyosin receptor kinase B (TrkB).[33,34] Although the endogenous mRNA level of BDNF in 1321N1 astrocytes seemed to be much higher than that of NGF, **3** and **2** slightly increased BDNF expression by about 1.9-fold (**3**; see Figure S3B) and 1.5-fold (**2**), respectively. The identity of both PCR products was verified by sequence analysis.

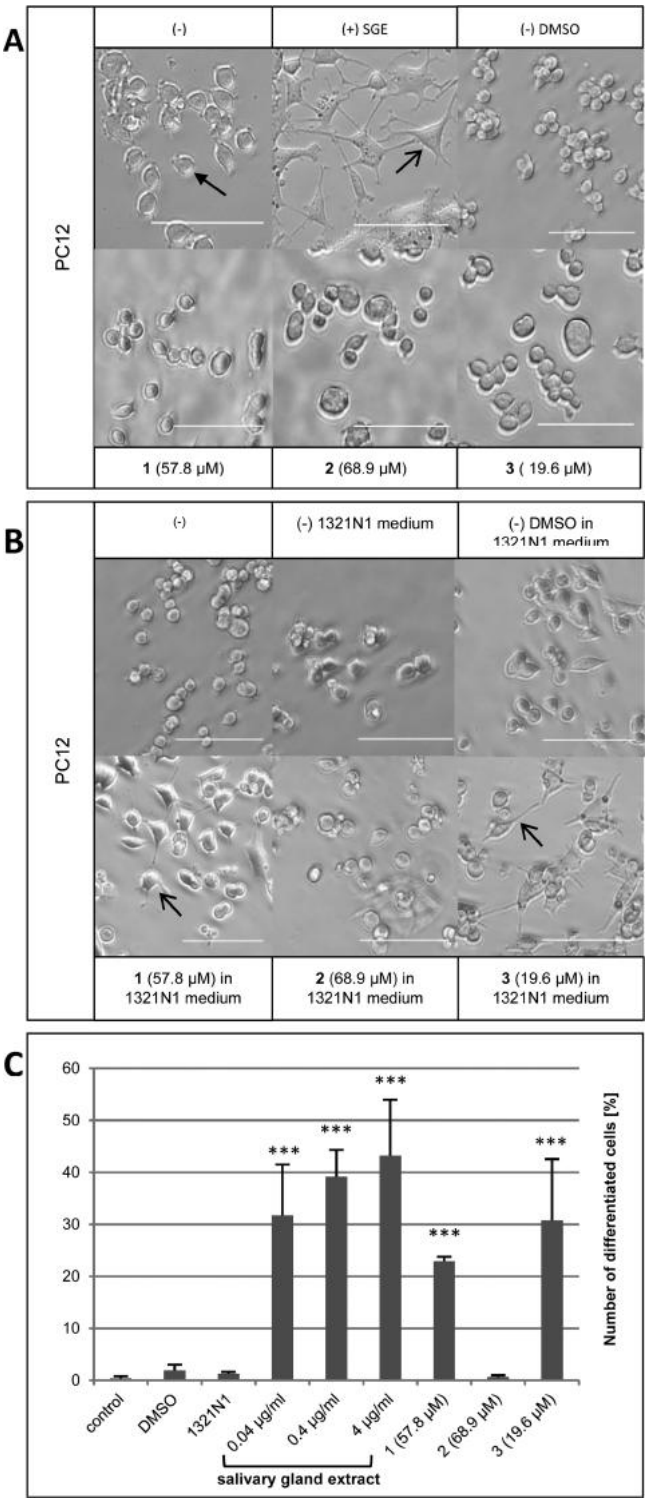

Figure 4. Morphological differentiation of PC12 cells incubated with corallocins (**1**−**3**) (A) or conditioned medium produced by 1321N1 cells (B). (−): negative control, no additive; (−) DMSO: negative control with DMSO; (−) 1321N1 medium: negative control with 1321N1 medium; (+) SGE: positive control with salivary gland extract (contains NGF). A nondifferentiated cell is marked with a filled arrow, whereas a differentiated cell is marked with an unfilled arrow. A quantification of this analysis is shown in C (±SEM; ***$p < 0.0001$).

Interestingly, **1** and **2** seem to act on 1321N1 cells differently by increasing transcription of either NGF or BDNF, respectively. Whereas **2** failed to induce neurite outgrowth

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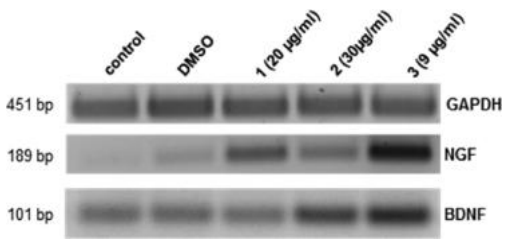

Figure 5. RT-PCR from 1321N1 astrocytes stimulated by corallocins A–C (**1**–**3**). Ubiquitously expressed GAPDH was used as loading control. PCR amplificates for NGF and BDNF were verified by sequencing (n = 3 independent experiments).

from cultured PC12 cells, it stimulated BDNF to a higher extent than **1**, which did not increase mRNA levels of BDNF in treated astrocytes. PC12 cells do not express the TrkB receptor[32] necessary to transduce the BDNF signal across the plasma membrane. This explains the lack of neurite outgrowth from PC12 cells incubated with corallocin B (**2**)-conditioned astrocyte medium. As compound **3** is able to stimulate the transcription of both neurotrophins, it may act upstream on a common molecular target of both pathways or via a third independent pathway.

In conclusion, three new compounds from *H. coralloides*, corallocins A–C, induce different patterns of neurotrophin expression in human astrocytes. For the first time we observed a promoting effect of fungal isoindolinone derivatives not just on NGF, but also on BDNF expression. Thus, they represent interesting tools for the investigation of neurotrophic systems.

■ EXPERIMENTAL PROCEDURES

General Experimental Procedures. 1D and 2D NMR spectra were recorded on a Bruker Avance III 700 spectrometer with a 5 mm TXI cryoprobe (^{1}H 700 MHz, ^{13}C 175 MHz) and a Bruker Avance III 500 (^{1}H 500 MHz, ^{13}C 125 MHz) spectrometer. IR spectra were measured in a PerkinElmer Spectrum 100 FTIR spectrometer, and UV spectra were recorded with a UV-2450 Shimadzu UV–vis spectrophotometer. All HPLC-MS analyses were performed on Agilent 1260 Infinity Systems with diode array detector and C_{18} Acquity UPLC BEH column (2.1 × 50 mm, 1.7 μm) from Waters. Solvent A: H_2O + 0.1% formic acid, solvent B: acetonitrile + 0.1% formic acid, gradient system: 5% B for 0.5 min increasing to 100% B in 19.5 min, maintaining 100% B for 5 min, flow rate = 0.6 mL min^{-1}, UV detection 200–600 nm. LC-ESIMS spectra were recorded on an ion trap MS (amaZon speed, Bruker), and HR-ESIMS spectra on a time-of-flight (TOF) MS (MaXis, Bruker). Chemicals and solvents were obtained from AppliChem GmbH, Avantor Performance Materials, Carl Roth GmbH & Co. KG and Merck KGaA in analytical and HPLC grade.

Fungal Material. Fresh fruiting bodies of *Hericium coralloides* were purchased from Pilzgarten GmbH, Helvesiek (Germany). A voucher specimen and a corresponding culture that showed the characteristics of the species are kept in the herbarium of the HZI Braunschweig, (Acc. No STMA 14303).

Extraction and Isolation. The basidiomes of *H. coralloides* (2.8 kg) were homogenized mechanically with an Ultra Turrax and stirred with acetone (4 L) for 48 h. After separation of the biomass by filtration the organic solvent was evaporated under reduced pressure (40 °C). The remaining aqueous residue was extracted with ethyl acetate twice (1 L). After drying over sodium sulfate, the organic solvent was removed under reduced pressure (40 °C) to yield the crude extract (7.88 g). For isolation of the compounds the extract was fractionated on silica (80 g) by flash chromatography (GRACE Reveleris X2 flash system) using a mixture of petroleum ether (solvent A) and ethyl acetate (solvent B); gradient: 0% to 70% (solvent B) in 35 min, 70% to 100% (B) in 5 min. Four fractions were collected: (1) 5% to 45% (B), 178 mg; (2) 46% to 70% (B), 3.38 g; (3) 76% to 99% (B), 821 mg; (4) 100% (B), 936 mg. Fraction 3 was subjected to preparative HPLC [Gilson GX270 Series HPLC system; column: VP Nucleodur 100-5 C_{18} ec (250 × 21 mm, 5 μm; Macherey-Nagel)] for further purification. A mixture of water (Milli-Q, Millipore, solvent A) and acetonitrile (HPLC-grade, solvent B) with 0.05% TFA was used as eluent with a flow rate of 20 mL/min. Gradient: 10% to 50% solvent B in 10 min, 50% to 85% B in 25 min, 85% to 100% B in 20 min, 100% B isocratic for 5 min. Corallocin A (**1**, 2.8 mg) was obtained at the retention time t_R = 23.5–24.5 min, corallocin B (**2**, 29.4 mg) at t_R = 34–36.5 min, corallocin C (**3**, 3.4 mg) at t_R = 40–41 min, and hericerin (**4**, 14.7 mg) at t_R = 42.5–43.5 min via UV detection at 210 and 280 nm. Fraction 4 was subjected to the same preparative HPLC system, and [5-(2*E*)-3′,7′-dimethyl-2′,6′-octadienyl]-4-hydroxy-6-methoxy-1-isoindoline (**5**, 2 mg) was obtained at t_R = 10.3–11.9 min using the following gradient: 58% B isocratic for 5 min, 58% to 70% B in 40 min, 70% to 100% B in 5 min.

*Corallocin A (**1**):* slightly yellow oil; IR (KBr) ν_{max} = 3410, 2960, 1685, 1480, 1445, 1390, 1350, 1205, 1140, 850, 800, 730; ^{1}H NMR and ^{13}C NMR data (500 MHz/125 MHz, acetone-d_6) see Table 1; HR-ESIMS m/z = 347.1490, calcd for $C_{19}H_{23}O_6$ $[M + H]^+$ m/z = 347.1489; ESIMS m/z (rel int) 715 (34) $[2M + Na]^+$, 693 (100) $[2M + H]^+$, 369 (6) $[M + Na]^+$, 347 (16) $[M + H]^+$, 329 (8) $[M - H_2O + H]^+$, 301 (4) $[M - CO_2H_2 + H]^+$.

*Corallocin B (**2**):* colorless oil; IR (KBr) ν_{max} = 3280, 2935, 2870, 1660, 1600, 1520, 1480, 1370, 1335, 1200, 1160, 1120, 1075, 840, 770; ^{1}H NMR and ^{13}C NMR data (500 MHz/125 MHz, acetone-d_6) see Table 1; HR-ESIMS m/z = 436.2484, calcd for $C_{27}H_{34}NO_4$ $[M + H]^+$ m/z = 436.2482; ESIMS m/z (rel int) 871 (100) $[2M + H]^+$, 458 (3) $[M + Na]^+$, 436 (55) $[M + H]^+$.

*Corallocin C (**3**):* colorless oil; IR (KBr) ν_{max} = 3410, 2935, 2870, 1675, 1480, 1335, 1210, 1150, 1120, 835, 800, 740, 730; ^{1}H NMR and ^{13}C NMR data (500 MHz/125 MHz, acetone-d_6) see Table 1; ^{1}H–^{15}N-HMBC (700 MHz, acetone-d_6) δ_N = 125.5 (N-5″), 121.8 (N-2), NH_3 was used for calibration; HR-ESIMS m/z = 459.2640, calcd for $C_{29}H_{35}N_2O_3$ $[M + H]^+$ m/z = 459.2642; ESIMS m/z (rel int) 917 (100) $[2M + H]^+$, 481 (3) $[M + Na]^+$, 459 (66) $[M + H]^+$.

Cytotoxicity Assay. In vitro cytotoxic effects (IC_{50}) against several mammalian cell lines were determined with 3.0 μL (30 μg/mL stock solutions in methanol) of the compounds by serial dilution (60 μL) in 96-well plates for tissue cultures (Falcon). The assay included mouse fibroblast cell line L929, breast cancer cell line MCF-7, human cervix carcinoma cell line KB-3-1, and umbilical vein endothelial cell line (HUVEC). Line L929 was cultured in Dulbecco's modified Eagle's medium (DMEM; Lonza), and MCF-7 and KB-3-1 were cultured in RPMI-1640 medium (Gibco), all supplemented with 10% fetal bovine serum (FCS; Gibco) and incubated under 5% CO_2 at 37 °C for 5 days. Methanol was used as negative control, and epothilone A as the positive control. The assays were conducted in accordance with literature descriptions.[30]

PC12 and 1321N1 Cell Stimulation Assays. 1321N1 astrocyte cells were cultivated in DMEM containing 10% FCS, penicillin (0.15 mM), streptomycin (86 μM), and glutamine (2 mM). PC12 cells were grown in DMEM supplemented with 5% FCS, 10% horse serum, penicillin (0.15 μM), streptomycin (86 μM), and glutamine (2 mM). Cells were incubated on 0.005% collagen-precoated plates in an incubator with 5% CO_2 at 37 °C. Collagen-coated 24-well plates were seeded with 7 × 10^4 PC12 cells/well. After 24 h of incubation *Hericium* compounds or increasing concentrations of NGF-containing salivary gland extracts were added to the medium, and cells were cultivated for an additional 48 h. PC12 cells were analyzed by transmitted light microscopy for neurite outgrowth.

The 1321N1 astrocyte cells were seeded on 3.5 cm plates and allowed to grow to 100% confluence. Twenty-four hours before adding compounds cells were treated with serum-free DMEM. After 48 h of incubation with *Hericium* compounds, the conditioned medium was collected and centrifuged to remove cells. Collagen-coated 24-well plates were seeded with 7 × 10^4 PC12 cells/well and incubated for 24 h. Subsequently, the 1321N1-conditioned medium was applied for 48 h followed by neurite outgrowth analysis. As negative control, PC12 cells were incubated with DMSO, and for the positive control the cells

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were treated with 4 μg/mL NGF-containing mouse salivary gland extract.

PC12 cells were analyzed by transmitted light microscopy for neurite outgrowth. Neurites longer than one cell diameter indicated differentiated cells. Three independent experiments were performed with approximately 100 cells analyzed each.

Semiquantitative RT-PCR. 1321N1 astrocytes were seeded and incubated as described above. Six hours after compound incubation, total RNA was isolated using PeqGold RNAPure (Peqlab). First-strand cDNA was synthesized from 1 μg of total RNA using AMV reverse transcriptase (Promega) and oligo(dT) primer (Promega). PCR primers for amplification and the sizes of respective PCR products were as follows: GAPDH (sense: 5′-TCCACCACCCTGTT-GCTGTA-3′; antisense: 5′-CCACAGTCCATGCCATCAC-3′; 451 bp), NGF (sense: 5′-CCAAGGGAGCAGTTTCTATCCTGG-3′; antisense: 5′-GCAGTTGTCAAGGGAATGCTGAAGTT-3′; 189 bp), BDNF (sense: 5′-TAACGGCGGCAGACAAAAAGA-3′; antisense: 5′-GAAGTATTGCTTCAGTTGGCCT-3′; 101 bp). PCR was carried out in a 25 μL volume containing cDNA template (2 μL), dNTP (10 mM, 0.5 μL), primers (100 μM; 0.1 μL), Go Taq buffer (5×, 5 μL), water (17.1 μL), and Go Taq polymerase (5 U/μL; 0.2 μL). The amplification programs were started with steps of 94 °C, 2 min and finished by 72 °C, 5 min and as follows: GAPDH, 20 cycles: 94 °C, 30 s, 60 °C, 30 s, 72 °C, 45 s; NGF, 30 cycles: 94 °C, 30 s, 61 °C, 30 s, 72 °C, 30 s; BDNF, 30 cycles: 94 °C, 30 s, 58 °C, 30 s, 72 °C, 30 s. GAPDH was used as loading control. The mRNA level of GAPDH, NGF, and BDNF was analyzed on a 1% or 2% agarose gel.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jnatprod.6b00371.

Analytical data of compounds **1**−**5**; 1D and 2D NMR, HR-ESIMS, and IR spectra of corallocins A−C; Figures S1, S2, and S3A/B2 (PDF)

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Notes

The authors declare no competing financial interest.

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